

Mand.ro Mark 7

Communication Protocol

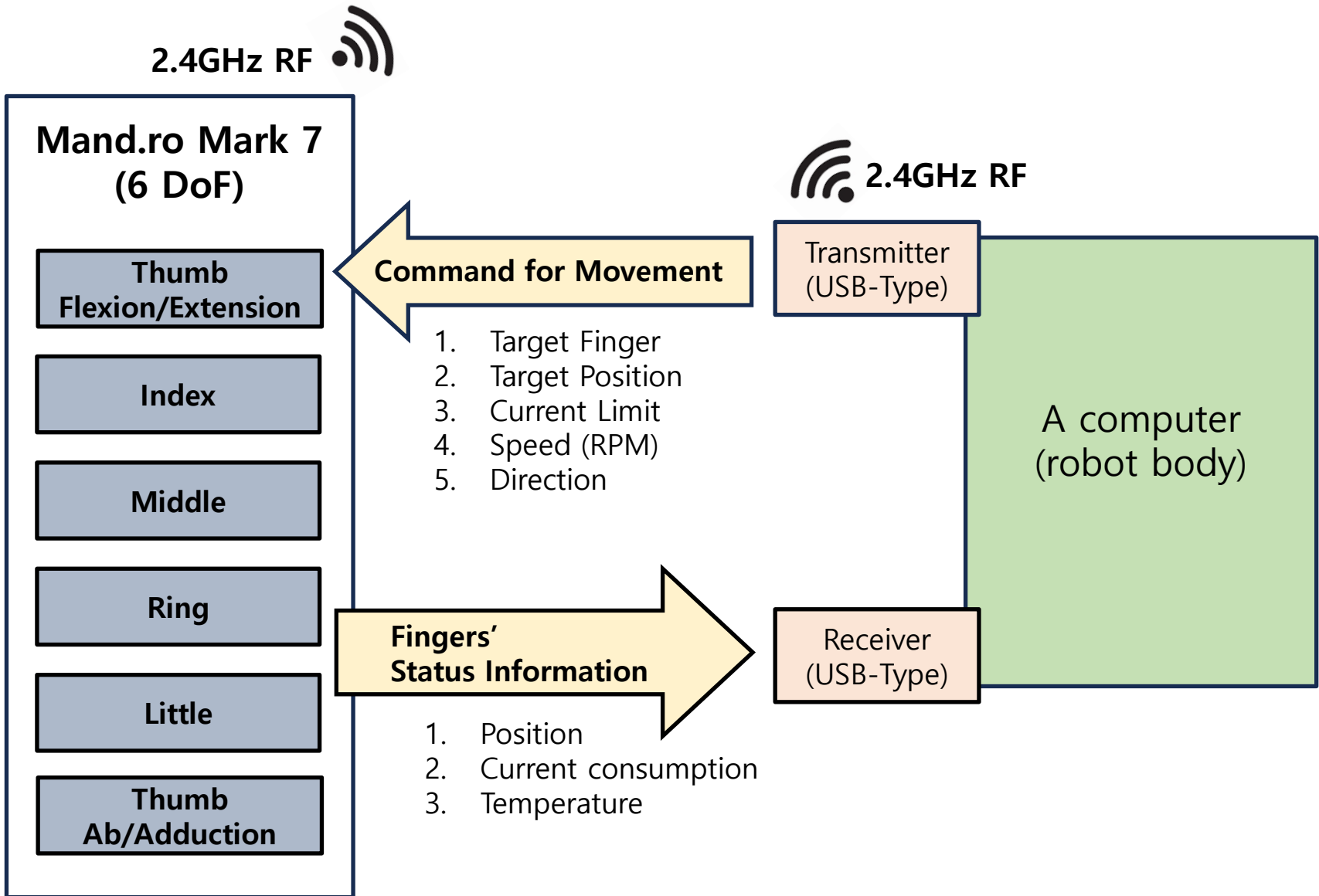
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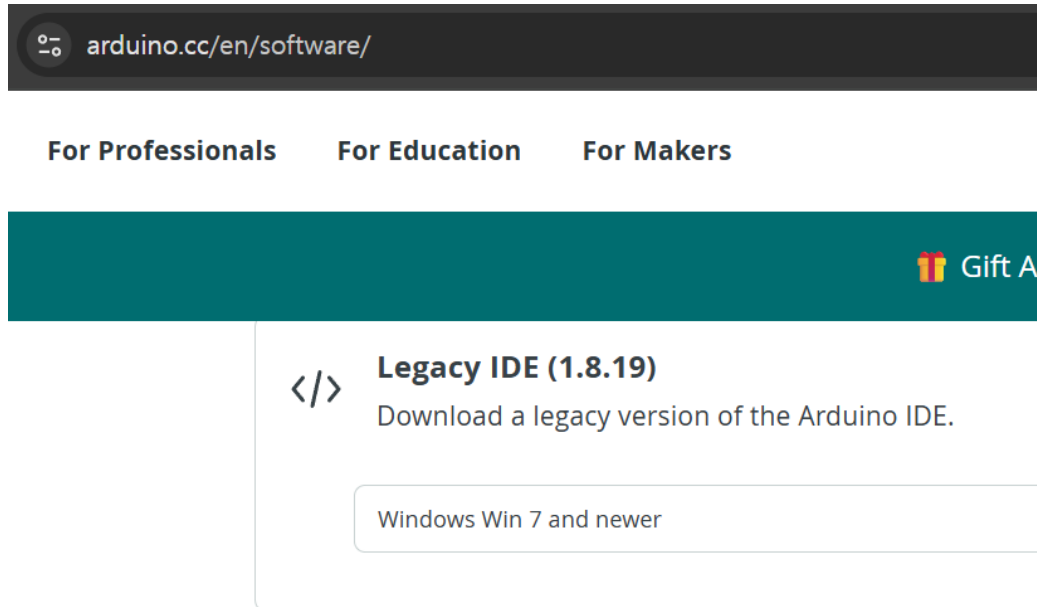
- Mand.ro Mark 7
- Note: Test Program – Comport Master

Mand.ro Mark 7

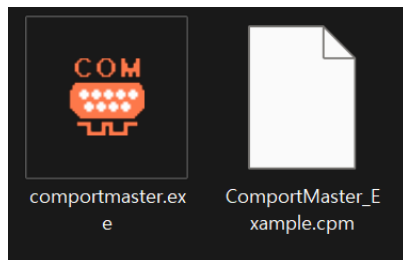


Prerequisites

- Install Arduino IDE (1.8.19 is recommended version)
 - This is required to install a USB-Serial Driver for Arduino-compatible boards.
 - Note that we use Arduino Micro board for our USB Type-A dongle for wireless communication.



- Install Comport Master program
 - This is a third-party program for easier Serial communication and testing.



PC to Hand Command Protocol (via Transmitter Dongle)

- Insert a Transmitter-Dongle into a PC, and Establish a Serial (COM port) Communication
 - Port Name: Arduino Pro Micro (COM xx)
 - Baud Rate: 115200 bps (8-N-1)
- Tx Packet Length: 15 bytes (15 == 2+6+2+2+2+1)
 - Header:2, Finger: 6, Speed:2, Current:2, Position:2, Direction: 1

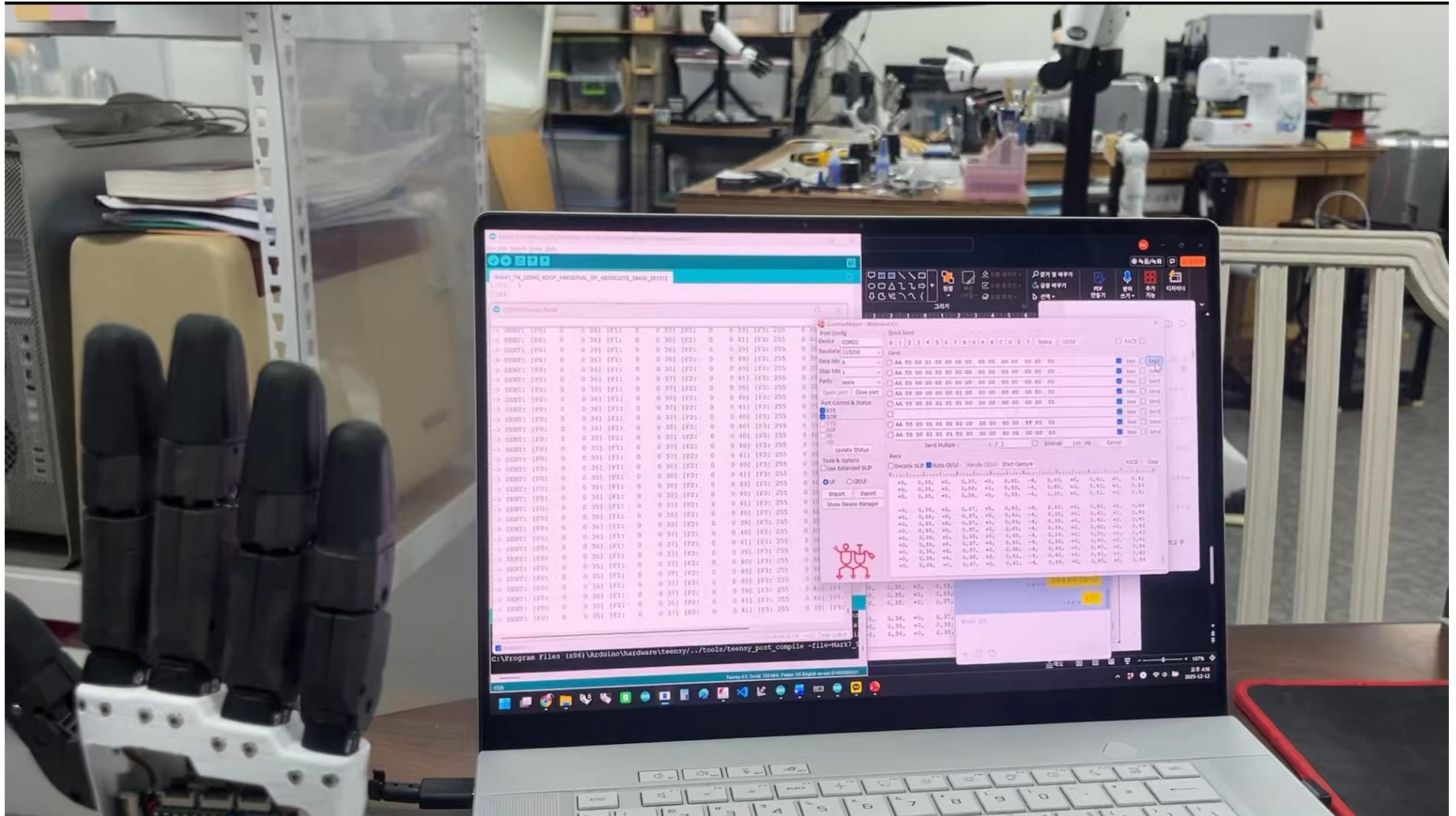
Byte Sequence	B0	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	B11	B12	B13	B14
Description	Header		Fingers (Thumb to Little, Thumb Abduction)						Speed		Current		Position		Direction
Data (Total 16 Bytes)	0xAA	0x55	0 or 1	0 or 1	0 or 1	0 or 1	0 or 1	0 or 1	High 8-bit	Low 8-bit	High 8-bit	Low 8-bit	High 8-bit	Low 8-bit	0, 1, 2, 3
Note	Binary Sequence		1: Selected, 0: Not Selected						RPM/10* (300 ~ 3000)		mA (600 ~ 1200)		Absolute from the initial position (-50 to 400)		0: Idle 1: Forward 2: Reverse 3: Reset
Extra Note			B2: Thumb Flexion B3: Index Flexion B4: Middle Flexion B5: Ring Flexion B6: Little Flexion B7: Thumb Abduction						ignored by 'zero' value • RPM/10: 300 means 3000 RPM		ignored by 'zero' value		ignored by 'zero' value		3: Reset Position Counter to Zero

Important Note for Absolute Position Control

- The absolute position is calculated from the initial finger position. (fully released)
- However, **the hall-effect sensors in the motor may have slips**, so it is recommended to reset its position counter when the hand is fully released while using.
- With the absolute position parameter, the 'forward direction' is the only one command to control the position. For example, if you make the following commands sequentially by time:
 - 1) Forward, Position Set as 100 -> Goes from zero to 100 position (grasping movement)
 - 2) Forward, Position Set as 50 -> Goes from 100 to 50 position (reverse action)
 - 3) Forward, Position Set as 200 -> Goes from 50 to 200 position (grasping movement)
 - 4) Forward, Position Set as **-10** -> Goes from 200 to **-10** position (reverse action)
 - * -10 is not reachable position, but the position counter has slips, so it is better to make below zero position setting for full releasing fingers
- **Note that setting zero (0) as a position value means it does not change the target position previously given to the finger.**
- The position parameter can be **reset to** zero value by **reset command (direction as 3).**

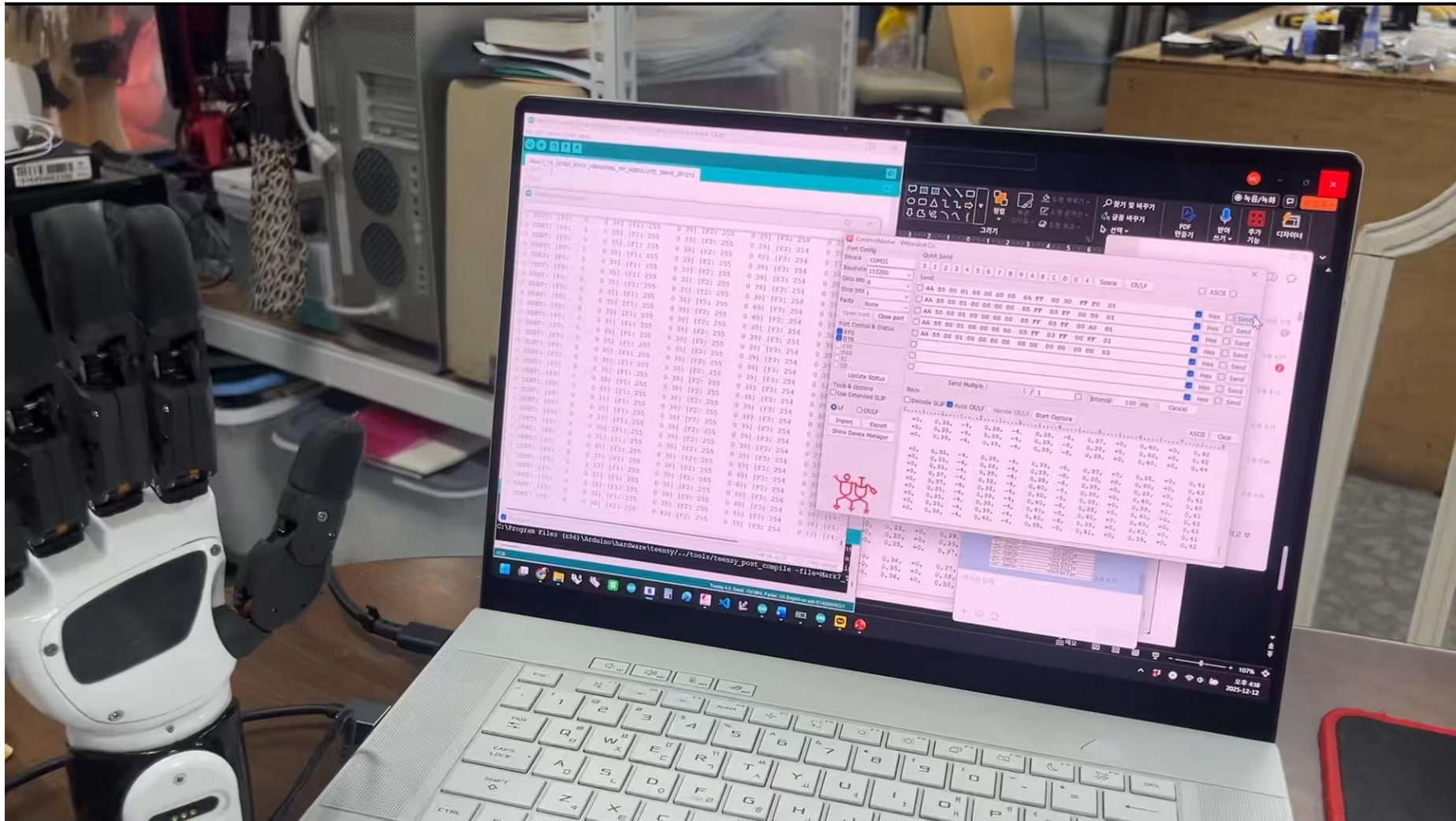
Example 1: Multiple Position Settings Among Fingers

- Initially set the target position per finger
- Make grasping (forward) direction command without 'zero' target position parameter



Example 2: Absolute Finger Position Control

- Use only grasping (forward) direction command with the 'target position' parameter.
 - When returning the finger to the released position, use a small negative position (0xFFFF0).



Example 3: Pipeting Example

- Initially grasp 3,4,5 fingers to hold a pipet, and use the thumb to press the handle.
 - Due to the possibility of signal loss over wireless communication, it is better to send 2 or 3 commands to make sure its delivery from USB Dongle to Mark 7 Hand.



Hand to PC Command Protocol (via Rx Dongle)

- Insert a Rx-Dongle into a PC, and Establish a Serial (COM port) Communication
 - Port Name: Arduino Pro Micro (COM xx)
 - Baud Rate: 115200 bps (8-N-1)
- Rx Packet Length: 19 bytes (19 == 3+3+3+3+3+3+1)
 - Thumb:3, Index:3, Middle:3, Ring:3, Little:3, Thumb Ab/Ad: 3, Checksum: 1

Byte Sequence	B0	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	B11	B12	B13	B14	B15	B16	B17	B18
Description	Position	Current (mA)	Temperature	Position	Current (mA)	Temperature	Position	Current (mA)	Temperature	Position	Current (mA)	Temperature	Position	Current (mA)	Temperature	Position	Current (mA)	Temperature	Checksum
Data (Total 19 Bytes)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	XOR (B0~B17)
Note	Divide by 4	Divide by 10	Celsius Degree	Divide by 4	Divide by 10	Celsius Degree	Divide by 4	Divide by 10	Celsius Degree	Divide by 4	Divide by 10	Celsius Degree	Divide by 4	Divide by 10	Celsius Degree	Divide by 4	Divide by 10	Celsius Degree	Validity Check
Extra Note	Thumb			Index			Middle			Ring			Little			Thumb Ab/Ad			

Note: Test Program: Comport Master (third-party program)

ComPortMaster - Withrobot Co.

Port Config

Device: COM52

Baudrate: 115200

Data bits: 8

Stop bits: 1

Parity: None

Open port Close port

Port Control & Status

☒ RTS

☒ DTR

☐ CTS

☐ DSR

☐ RI

☐ CD

Update Status

Tools & Options

☐ Use Extended SLIP

☒ LF ☐ CR/LF

Import Export

Show Device Manager

Quick Send

0 1 2 3 4 5 6 7 8 9 A B C D E F Space CR/LF

☐ ASCII

Send

☐	AA 55 01 00 00 00 00 00 05 FF 04 30 00 A0 01	☒	Hex	☐ Send
☐	AA 55 01 00 00 00 00 00 05 FF 04 30 FF F0 01	☒	Hex	☐ Send
☐		☒	Hex	☐ Send
☐	AA 55 00 00 01 01 01 00 0A FF 04 30 01 70 01	☒	Hex	☐ Send
☐	AA 55 00 00 01 01 01 00 0A FF 04 30 FF F0 01	☒	Hex	☐ Send
☐	AA 55 01 01 01 01 01 01 00 00 00 00 00 03	☒	Hex	☐ Send
☐	AA 55 00 01 00 00 00 01 05 FF 04 30 00 A0 01	☒	Hex	☐ Send
☐	AA 55 00 01 00 00 00 01 05 FF 04 30 FF F0 01	☒	Hex	☐ Send

Send Multiple : 1 / 1 Interval 100 ms Cancel

Recv

☐ Decode SLIP ☒ Auto CR/LF ☐ Handle CR/LF Start Capture

ASCII Clear

0... | ...1... | ...2... | ...3... | ...4... | ...5... | ...6... | ...7... | ...8

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